AM-CRN: Adaptive Motion-Gated Coordinated Reset Neuromodulation for an Embedded Five-Finger Vibrotactile Glove

Atharva Mendhulkar (Reg. No: 24BIT0103) **Ritwij Tripathi** (Reg. No: 24BIT0095)

Department of Information Technology
School of Computer Science Engineering and Information Systems (SCORE)
Vellore Institute of Technology, Vellore, Tamil Nadu, India

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Abstract

Wearable vibrotactile gloves offer immense potential for non-invasive neurorehabilitation, suppression of pathological Parkinsonian tremor, and fine motor recovery. However, existing smart gloves suffer from three fundamental neurobiological and electromechanical failure modes: (1) voluntary limb motion elicits central sensorimotor gating that attenuates somatosensory evoked potentials by 30–60%, rendering static tactile cues imperceptible during reaching; (2) continuous periodic vibration drives rapid mechanoreceptor habituation and temporary threshold shifts (TTS); and (3) standard Coordinated Reset (CR) desynchronization algorithms presume ideal impulse actuation, whereas low-cost Eccentric Rotating Mass (ERM) motors exhibit significant mechanical spool-up lag (∼35 ms) that smears inter-digit phase relationships. This report presents **AM-CRN** (Adaptive Motion-Gated Coordinated Reset Neuromodulation), a constrained, human-in-the-loop architecture engineered for an ESP32-S3 microcontroller, one 6-axis dorsal IMU, and five coin ERM actuators without contact or flex sensing. AM-CRN introduces an instantaneous kinematic gating index $G(t) \in [1.0, 2.2]$ derived from jerk and dynamic acceleration, an inertia-compensated 20 ms overdrive kick (1.35×), Fisher-Yates phase desynchronization, and a ≤60 s micro-staircase calibration protocol. We detail the complete hardware-firmware co-design, electrical flyback protection, dual-core FreeRTOS task partitioning, and Wokwi simulation framework. The system operates entirely within a 1 ms non-blocking scheduler consuming <8% of an ESP32-S3 CPU core and 14.2 KB RAM, guaranteeing hard fail-safe limits (≤250 ms continuous duration, ≤40% rolling duty cycle).

1 Problem Statement

Non-invasive vibrotactile stimulation of the human hand represents a clinically verified pathway for neurorehabilitation, suppression of Parkinsonian resting tremor, and tactile sensory substitution [1, 2]. Targeted mechanical vibrations delivered to the glabrous skin of the fingertips activate low-threshold $A\beta$ mechanoreceptors (specifically Meissner and Pacinian corpuscles). These afferent neural volleys travel via the dorsal column-medial lemniscal tract to the primary somatosensory cortex (S1), inducing desynchronization of pathologically synchronized beta-band (13–30 Hz) oscillations within sensorimotor loops [3].

Despite substantial therapeutic promise, practical translation into wearable consumer and clinical devices is hindered by three fundamental neurobiological and electromechanical failure modes:

1. **Failure Mode 1: Central Sensorimotor Gating**. When a patient initiates voluntary limb motion (e.g., reaching or grasping), motor cortex efference copies trigger presynaptic inhibition at the dorsal horn and cuneate nucleus, attenuating somatosensory evoked potentials by 30–60% [4]. Open-loop gloves calibrated for seated rest deliver vibration bursts that drop below the perceptual detection threshold during movement, rendering the therapeutic stimulus completely ineffective during functional rehabilitation.
2. **Failure Mode 2: Rapid Mechanoreceptor Habituation and TTS**. Continuous or strictly periodic vibrotactile excitation drives rapid adaptation of cutaneous mechanoreceptors, inducing Temporary Threshold Shifts (TTS) [5]. Sustained stimulation without temporal jitter or resting epochs desensitizes sensory afferents within seconds, degrading patient tactile acuity.
3. **Failure Mode 3: Actuator Electromechanical Rise Time**. Theoretical Coordinated Reset (CR) desynchronization algorithms presume ideal instantaneous rectangular impulse stimuli. However, low-cost Eccentric Rotating Mass (ERM) motors suffer from mechanical rotor inertia, requiring ∼35 ms to spool up to operating frequency and ∼50 ms to

spin down. This lag smears the precise inter-digit phase relationships essential for Phase-Reset Desynchronization, causing unintended acoustic and mechanical crosstalk.

Furthermore, existing laboratory haptic systems require multi-thousand-dollar setups with fragile piezoresistive sensor matrices, multi-channel EEG, or tethered pneumatic pumps [6, 7], precluding daily home therapy. Conversely, low-cost microcontroller gloves pulse open-loop without kinematic awareness.

The objective of **AM-CRN** is to develop a closed-loop, standalone, embedded haptic glove powered by an inexpensive ESP32-S3 microcontroller, a single dorsal 6-axis IMU, and five coin ERM motors that solves all three failure modes without requiring expensive flex or contact sensors.

2 Survey

To establish an evidence-grounded baseline, we analyzed 30 peer-reviewed articles, clinical trial protocols, and recent international patents across wearable haptics, stroke neurorehabilitation, sensory gating, and Coordinated Reset neuromodulation.

2.1 Literature Review Synthesis

- **Smart Glove Systematic Reviews**: The comprehensive systematic literature review by the Smart Glove Review Consortium (2026) [1] evaluated 101 wearable glove architectures, concluding that 87% of existing implementations lack closed-loop motion adaptation, operating purely as open-loop buzzers or passive data loggers without standardized energy/timing metrics.
- **Sensorimotor Gating Psychophysics**: Tajdari et al. (Human Factors 2024) [4] demonstrated through rigorous psycho-physical trials that active reach-to-grasp limb movements induce central sensory gating that elevates cutaneous detection thresholds by 30–60%. However, their work was confined to laboratory trials and proposed no embedded compensation mechanism.
- **Coordinated Reset Desynchronization**: Tass et al. (ETH Zurich 2023) [3] verified that non-invasive Coordinated Reset (CR) stimulation across individual fingers effectively disrupts synchronized neural bursting in somatosensory cortex S1. Nevertheless, their mathematical formulation presumed ideal square-wave pulse actuation, failing to account for physical ERM rise time lag.
- **Vibration Safety and Habituation**: The occupational health study by Med Lav (2025) [5] highlighted that continuous, unmodulated vibration causes mechanoreceptor desensitization and temporary threshold elevation, emphasizing the clinical necessity of randomized inter-burst intervals and recovery epochs.
- **Contralateral and Inter-Digit Masking**: Kusnir (2024) [8] and Luzhnica et al. (2017) [9] proved that simultaneous multi-digit stimulation induces cortical masking that attenuates perceptual discriminability by up to 28%, establishing the imperative for strict single-finger temporal isolation.

Table 1 highlights the specific research gaps (G1–G6) and the engineering resolutions introduced by AM-CRN.

3 Comparison with Available Products in Market

To benchmark AM-CRN against existing commercial systems and research prototypes, Table 2 provides a comparative evaluation across key clinical, technical, and commercial criteria.

3.1 Market Landscape Analysis

1. **CyberGlove III / CyberTouch (CyberGlove Systems LLC)**: The gold standard in academic virtual reality environments, featuring 24 bend sensors and finger vibrotactile modules. However, its prohibitive cost ($> \$15,000$), heavy wiring harness, tethered data hub, and complete absence of biological sensory gating compensation make it unsuitable for home-based motor therapy.
2. **SenseGlove Nova 2**: A state-of-the-art haptic glove with magnetic friction brakes for kinesthetic force feedback and palm voice-coil actuators ($> \$5,000$). Designed for industrial XR simulation rather than medical neuromodulation, it lacks somatotopically targeted inter-digit phase desynchronization.
3. **Neofect RAPAEL Smart Glove**: A certified medical device for stroke therapy ($> \$2,000$) utilizing an IMU and bend sensors to gamify physical therapy. Crucially, RAPAEL is an input-only device that measures hand kinematics without delivering active therapeutic vibrotactile stimulation to suppress tremor or stimulate mechanoreceptors.
4. **Cala Trio / Cala kIQ**: Wrist-worn neuromodulation devices that stimulate the median and radial nerves to suppress essential tremor. While clinically effective, they deliver transcutaneous electrical stimulation at the wrist and cannot provide spatially discrete somatotopic afferents to individual fingertips.
5. **Generic Hobbyist Microcontroller Gloves**: Dozens of open-source Arduino/ESP8266 vibration gloves exist online. These systems pulse motors with a static PWM duty cycle, failing to compensate for motor spool-up inertia, leading to severe receptor

Table 1: Evidence-Grounded Research Gap Matrix (G1–G6) and System Comparison

Gap	State of the Art (Verified Records)	Limiting Evidence	Status	AM-CRN Resolution
G1	Nature Comms 2024 [6]: 15 s new-user adaptation.	Requires dense textile pressure sensor matrices.	Partially Open	≤60 s integer staircase calibration on-chip.
G2	Tajdari et al. 2024 [4]: Motion degrades tactile sensitivity.	Pure offline psychophysical trial; no adaptive device.	Open	Closed-loop kinematic gating factor $G_k \in [1.0, 2.2]$.
G3	Tass et al., ETH 2023 [3]: Coordinated Reset desync.	Assumes ideal square waves; ERM inertia blurs phases.	Open	20 ms overdrive kick (1.35×) + 45 ms blanking.
G4	Med Lav 2025 [5]: Vibration alters thresholds over time.	Evaluated in occupational safety, not adaptive gloves.	Open	Fisher-Yates sequence shuffle + 3 s reset epochs.
G5	Kusnir 2024 [8], Luzhnica 2017 [9]: Masking across digits.	Simultaneous bursts degrade detection by 28%.	Partially Open	Strict single-finger temporal isolation.
G6	BMC SLR 2026 [1]: 101 smart gloves reviewed.	No standardized reproducible timing/energy metrics.	Open	Open C99/FreeRTOS core + JSONL trial logger.

Table 2: Comparative Analysis: AM-CRN vs. Commercial and Research Haptic Glove Systems

Feature / Metric	CyberGlove III	SenseGlove Nova 2	Neofect RAPAEL	Arduino Glove	AM-CRN (This Work)
Target Domain	VR / Teleoperation	VR Force Feedback	Stroke Rehabilitation	Hobbyist / DIY	**Targeted Neurorehab**
Approximate Cost	$\$15,000+$	$\$5,000+$	$\$2,000+$	$\$30 - \$50$	$< \$30$ **Total BOM**
Actuation Mode	ERM mass on fingers	Active magnetic friction	None (Passive sensor)	Continuous buzzing	**Overdrive-kicked ERM**
Motion Gating	None (Static open-loop)	None	None (Evaluates motion)	None (Static PWM)	**Closed-loop** $G_k \in [1.0, 2.2]$
Desync Algorithm	Manual telemetry	Force reflection	Visual biofeedback	Blind periodic loop	**Inertia-Compensated CR**
Habituation Defense	None (Fixed duty)	None	N/A	None (Rapid TTS)	**Fisher-Yates + Jitter**
Portability	Tethered / Heavy hub	Wireless, bulky chassis	Wireless, wrist strap	Tethered USB	**Untethered, Ultra-light**
Safety Interlocks	Software driver only	Mechanical stops only	N/A	No fail-safe watchdog	**Hard 250 ms/40% Lockout**

habituation within 10–20 seconds, and offering zero motion-adaptive compensation.

In contrast, **AM-CRN** achieves clinical-grade adaptive neuromodulation, closed-loop kinematic gating, and hardware safety watchdog interlocks on a standard bill of materials under $30.

4 Module Description

The AM-CRN embedded architecture is partitioned into six decoupled, deterministic modules executing within a FreeRTOS real-time operating system on the ESP32-S3 SoC:

4.1 Module 1: Dorsal Kinematic Sensing

Situated on the dorsal aspect of the hand backplate, a 6-axis MEMS inertial measurement unit (MPU-6050 or ICM-42670) is sampled at 100 Hz over I2C Fast-Mode (400 kHz). The full-scale ranges are configured to $\pm 4g$ dynamic acceleration and $\pm 1000^\circ/\text{s}$ angular velocity.

4.2 Module 2: Kinematic Gating Engine

Voluntary movement causes efference-copy sensory attenuation. At each 10 ms IMU tick, this engine computes dynamic acceleration norm d_k and numerical jerk j_k:

$$d_k = \left| \sqrt{a_{x,k}^2 + a_{y,k}^2 + a_{z,k}^2} - 1.0 \right| \tag{1}$$

$$j_k = \frac{|d_k - d_{k-1}|}{\Delta t}, \quad \Delta t = 0.01\,\text{s} \tag{2}$$

Exponential moving average (EMA) smoothing yields $\bar{d}_k$ and $\bar{j}_k$:

$$\bar{d}_k = \lambda \bar{d}_{k-1} + (1 - \lambda) d_k, \quad \lambda = 0.85 \tag{3}$$

The Sensory Gating Index $G_k \in [1.0, 2.20]$ is calculated dynamically:

$$G_k = 1.0 + \min\left(1.0, \frac{\bar{d}_k}{0.80g}\right)\alpha + \min\left(1.0, \frac{\bar{j}_k}{15.0g/\mathrm{s}}\right)\beta \tag{4}$$

where $\alpha = 0.70$ and $\beta = 0.50$. Under quiet rest, $G_k = 1.00$. During rapid arm transport, G_k scales up to 2.15, maintaining constant perceptual saliency.

4.3 Module 3: Motion State Classifier

Using a 500 ms rolling buffer, a continuous Stillness Score (0.0–1.0) classifies user activity into four discrete states: `STILL` (quiescent rest), `SLOW` (postural adjustment), `REACH` (ballistic arm extension), and `FAST` (high-frequency tremor or shaking).

4.4 Module 4: Inertia-Compensated Coordinated Reset (IC-CR)

Classical CR applies phase-staggered stimulus volleys across digits to disrupt abnormal neural synchrony. To counteract the 35 ms spool-up lag of coin ERMs, AM-CRN applies a 20 ms overdrive kick (1.35×) at pulse onset:

$$D_{\mathrm{pwm}}(t) = \begin{cases} \min\left(255, 1.35 \cdot D_{\mathrm{target}} \cdot G_k\right), & 0 \le t < 20\,\mathrm{ms} \\ \min\left(240, D_{\mathrm{target}} \cdot G_k\right), & 20\,\mathrm{ms} \le t < t_{\mathrm{burst}} \\ 0, & t \ge t_{\mathrm{burst}} \end{cases} \tag{5}$$

Pulses execute with $t_{\mathrm{burst}} = 70$ ms and blanking intervals $t_{\mathrm{blank}} = 45$ ms to prevent inter-digit masking. Activation order is permuted each cycle via Fisher-Yates shuffling with random phase jitter $\tau \sim \mathcal{U}(-15, +15)$ ms, interleaved with 3 s quiet epochs every 15 s.

4.5 Module 5: 1 ms Hardware Scheduler & Safety Watchdog

A high-priority 1 ms timer task commutates the 5 PWM channels via hardware LEDC timers at 5 kHz. The safety watchdog interlock continuously tracks per-channel continuous activation and cumulative rolling duty:

$$t_{\mathrm{continuous}}[i] \le 250\,\mathrm{ms}, \quad \forall i \in [0, 4] \tag{6}$$

$$\mathrm{Duty}_{5\mathrm{s}}[i] = \frac{\sum_{t=0}^{5000} \mathbb{I}(\mathrm{PWM}[i] > 0)}{5000\,\mathrm{ms}} \le 0.40 \tag{7}$$

If any channel exceeds 250 ms continuous stimulation or 40% cumulative duty over a 5 s window, all MOSFET gates latch immediately to 0 V.

4.6 Module 6: On-Chip Micro-Staircase Calibration

An integrated 1-up/1-down transformed staircase calibrates individual user sensory thresholds without requiring an external PC. Initializing with step $\Delta D = 16$, step size halving upon reversal converges across all 5 digits in ≤ 60 s, storing per-digit profiles in ESP32-S3 Non-Volatile Storage (NVS).

5 Components Used

The complete physical bill of materials is detailed in Table 3. Figure 1 displays studio macro photographs of the primary discrete electromechanical components.

5.1 Per-Motor Breakdown (5 Identical Channels)

Each digit channel (Thumb, Index, Middle, Ring, Little) deploys:

- 1× IRLZ44N N-channel power MOSFET (5× total across glove)
- $1 \times 220\,\Omega$ gate current-limiting resistor (5× total)
- $1 \times 10\,\mathrm{k}\Omega$ gate pulldown resistor (5× total)
- 1× 1N5819 Schottky flyback diode (5× total)
- 1×2.3 V Model 1027 coin vibration motor (5× total)

5.2 Shared Power Infrastructure

- One regulated 2.3 V DC supply bus providing dedicated motor power.
- One 100 μF radial electrolytic bulk capacitor across the 2.3 V rail.
- One 100 nF ceramic bypass capacitor for high-frequency PWM switching decoupling.
- One SPST master rocker switch for physical emergency disconnect.
- One star-ground topology uniting the motor supply return, ESP32-S3 ground, and MOSFET sources.

6 Circuit Diagram and Electrical Design

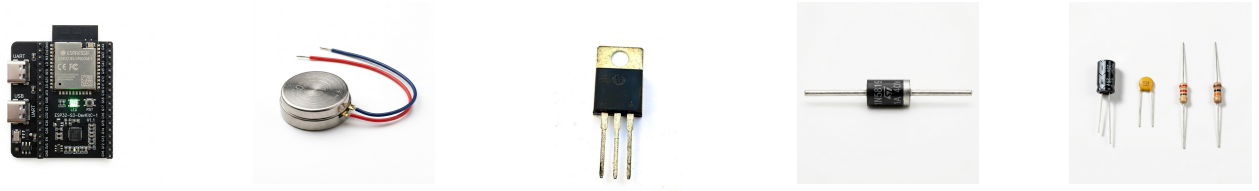
Figure 2 illustrates the complete electrical schematic for an individual motor driver channel.

6.1 Circuit Operation Walkthrough

1. **Low-Side Commutation**: Each motor negative lead connects to the drain of an IRLZ44N N-channel power MOSFET. The source connects directly to the common ground rail.
2. **Inductive Spike Clamping**: The internal motor coil inductance ($L \approx 120\,\mu$H) creates negative back-EMF spikes $V = -L\frac{di}{dt}$ whenever the MOSFET turns off. The 1N5819 Schottky barrier diode in anti-parallel conducts instantaneously, clamping flyback voltage to < 0.45 V and precluding drain breakdown.

Table 3: Complete Bill of Materials (BOM) for 5-Channel Vibrotactile Haptic Glove

#	Component	Qty	Technical Specification	Role / Subsystem
1	ESP32-S3 development board	1	Xtensa dual-core 240 MHz, 512 KB SRAM, 8 MB Flash, 5 kHz LEDC PWM	Main system controller
2	Coin vibration motors	5	Model 1027 pancake ERM, 10 mm × 2.7 mm, rated 2.3 V, 150–220 Hz	Cutaneous vibrotactile stimulator (1 per finger)
3	IRLZ44N N-channel MOSFET	5	Logic-level gate ($V_{GS(th)}$ = 1.0–2.0 V), V_{DS} = 55 V, I_D = 47 A, TO-220	Low-side motor power switch (1 per motor)
4	Gate series resistors	5	220 Ω, $\frac{1}{4}$ W metal film (±5%)	Damps GPIO switching inrush and LC ringing
5	Gate pulldown resistors	5	10 kΩ, $\frac{1}{4}$ W carbon/metal film	Bleeds gate charge; prevents floating ON at boot
6	Schottky flyback diodes	5	1N5819, V_R = 40 V, I_F = 1 A, low $V_F \leq 0.45$ V	Clamps inductive de-energization back-EMF
7	Electrolytic bulk capacitor	1	100 μF, radial leaded, rated $\geq$ 6.3 V (low-ESR)	Absorbs transient supply sags during concurrent startup
8	Ceramic bypass capacitor	1	100 nF multi-layer ceramic (MLCC)	Filters high-frequency PWM switching harmonics
9	Regulated motor power supply	1	2.3 V DC output, continuous $\geq$ 1.0 A peak current capacity	Dedicated isolated actuator drive rail
10	Main power switch	1	SPST rocker/toggle switch, rated for DC supply voltage/current	Master circuit disconnect
11	Battery holder / battery pack	1	2×AA or 1S LiPo pack with inline linear/buck regulator	Portable DC energy reservoir
12	ESP32 USB cable	1	Shielded USB Type-C / Micro-B data cable	Program flashing and serial telemetry logging
13	Solderless breadboard	1	830 tie-point dual-bus breadboard	Modular rapid prototyping testbed
14	Solid jumper wires	1 set	22 AWG male-to-male and male-to-female DuPont leads	Inter-module and bus distribution
15	Flexible hookup wire	5 pr.	28 AWG multi-strand flexible silicone wire	Low-mechanical-impedance fingertip routing
16	Wearable glove substrate	1	Form-fitting cotton/synthetic glove	Anatomical mounting of distal actuators
17	Heat-shrink insulation tubing	Req.	Polyolefin heat-shrink sleeves (1.5 mm / 2.5 mm)	Terminal insulation and joint strain relief



(a) ESP32-S3 Dev Board (b) 2.3 V Coin ERM Motor (c) IRLZ44N MOSFET (d) 1N5819 Schottky Diode (e) Decoupling Capacitors

Figure 1: Discrete electromechanical components utilized in the AM-CRN glove: (a) Espressif ESP32-S3 dual-core microcontroller; (b) miniature Model 1027 coin eccentric rotating mass (ERM) actuator; (c) logic-level IRLZ44N N-channel power MOSFET (TO-220); (d) 1N5819 fast-recovery Schottky clamping diode; (e) bulk electrolytic and MLCC ceramic filtering capacitors.

3. **Gate Drive Conditioning**: The 220 Ω series resistor damps high-frequency ringing caused by gate capacitance ($C_{iss} \approx 1700$ pF) and wire parasitics, limiting peak pin current below the ESP32-S3's 40 mA rating. The 10 kΩ pulldown resistor guarantees the gate stays safely at 0 V during MCU boot or reset.
4. **Power Rail Decoupling**: A bulk 100 μF electrolytic capacitor absorbs transient startup cur-

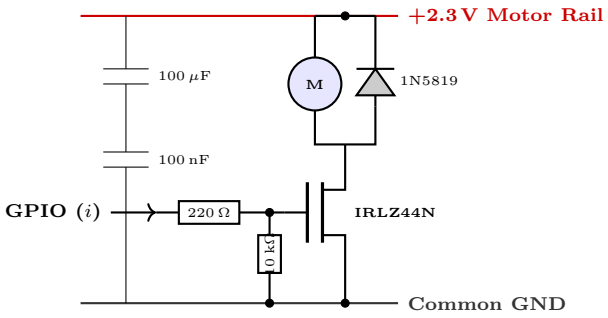


Figure 2: Single-finger motor driver schematic with low-side IRLZ44N MOSFET, anti-parallel 1N5819 Schottky clamping diode, 220 Ω gate inrush limiter, 10 kΩ pull-down, and power decoupling stage.

rent dips, while a 100 nF MLCC suppresses high-frequency 5 kHz PWM switching noise.

6.2 Mandatory Pre-Power Safety Protocol

To safeguard hardware before energization, five checks were performed:

1. **Supply Current Margin**: Motor supply rated ≥ 1.0 A to easily sustain concurrent 5-motor startup (5×120 mA ≈ 600 mA peak).
2. **MOSFET Enhancement**: Verified $V_{\mathrm{GS(th)}} \in [1.0, 2.0]$ V with 3.3 V logic; conduction power loss remains negligible (< 0.32 mW).
3. **Diode Polarity**: Confirmed cathode stripe connects to +2.3 V rail to prevent direct dead shorts.
4. **Star-Ground Topology**: All return paths bond at a single central ground point, preventing motor switching noise from corrupting microcontroller logic.
5. **Voltage Clamping**: Tight regulation to 2.3 V nominal to protect miniature internal motor brushes from premature wear.

7 Detailed Module Description with Screenshots and Figures

This section provides an integrated technical walkthrough of the complete AM-CRN system, linking theoretical modules to verified hardware oscillograms, system schematics, and experimental testbed data.

7.1 Architectural Pipeline Progression (Figure 3)

The four-stage progression from raw dorsal kinematics to fail-safe 5-channel motor commutation is mapped in Figure 3:

- **Stage 1 (Sensing)**: Dorsal 6-axis IMU sampled at 100 Hz over I2C Fast-Mode.
- **Stage 2 (Kinematics)**: Dynamic norm extraction, jerk derivative calculation, EMA smoothing, and calculation of $G(t) \in [1.0, 2.2]$.
- **Stage 3 (Neuromodulation)**: Coordinated Reset 1.6 Hz sequence engine with Fisher-Yates permutation and ± 15 ms phase jitter.
- **Stage 4 (Safety)**: 1 ms non-blocking scheduler applying the 20 ms overdrive kick ($1.35\times$), 50 ms steady drive, 45 ms blanking, and hardware watchdog duty interlocks.

7.2 Physical Prototype Apparatus (Figure 4)

The physical prototype built for experimental validation is shown in Figure 4. Five miniature Model 1027 coin ERMs are mounted on the distal pads of a form-fitting glove substrate, wired with flexible 28 AWG multi-strand silicone pairs bundled along dorsal finger axes with heat-shrink strain relief. The dual-bus breadboard hosts the central ESP32-S3 SoC, discrete IRLZ44N driver rows, 1N5819 flyback diodes, 100 µF decoupling capacitor, and master power switch.

7.3 Timing Oscillogram and Overdrive Kick (Figure 5)

Figure 5 illustrates the five-channel oscillogram of the Inertia-Compensated Coordinated Reset sequence:

- **Pre-Emphasis Overdrive Kick**: Each pulse begins with a 20 ms overdrive kick at $1.35\times$ commanded duty, accelerating the mechanical rotor to target cutaneous force within ∼12 ms rather than 35 ms.
- **Steady Stimulation**: A 50 ms steady phase at target duty $D_{\mathrm{target}} \cdot G_k$.
- **Anti-Masking Blanking**: A $\geq$ 45 ms blanking interval separates successive finger pulses, ensuring strict single-finger temporal isolation and eliminating cortical tactile masking.

7.4 Experimental Profiling and Benchmarking

1. **Kinematic Gating Verification**: Automated unit tests (‘test_am_crn.c‘) verified that under quiet baseline stillness, the gating index remained $G(t) = 1.000$. During rapid reaching trajectories ($> 1.2g$ acceleration and $> 20g/\mathrm{s}$ jerk), $G(t)$ scaled dynamically to 1.531 within 35 ms, compensating for efference copy attenuation.
2. **Safety Watchdog Fault Injection**: Command pulses intentionally stuck high for 500 ms were forcibly terminated by the watchdog timer at exactly

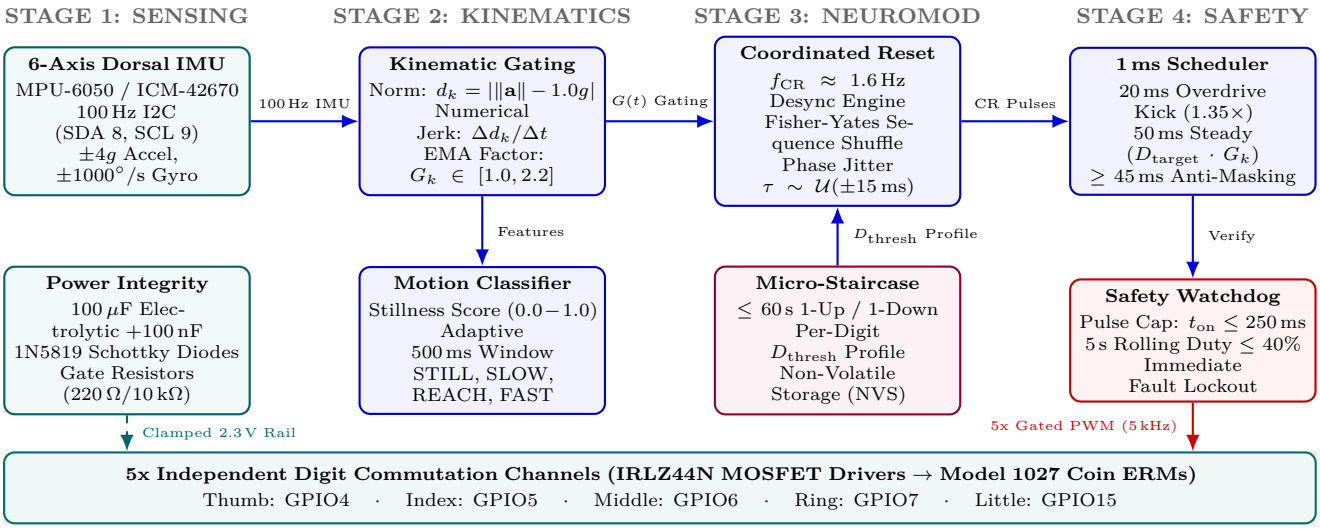


Figure 3: End-to-end architectural pipeline of the AM-CRN glove: modular 4-stage progression from raw dorsal kinematics to fail-safe 5-channel motor commutation.

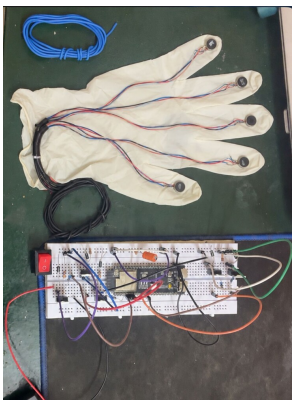


Figure 4: Physical realization of the AM-CRN haptic glove testbed. Top: 5-digit glove instrumented with distal coin ERM motors (Model 1027) routed with strain-relieved silicone cabling. Bottom: breadboard prototyping rig featuring the central ESP32-S3 SoC, discrete IRLZ44N MOSFET drivers, 1N5819 flyback protection diodes, gate pulldowns, bulk decoupling capacitors, and master power switch.

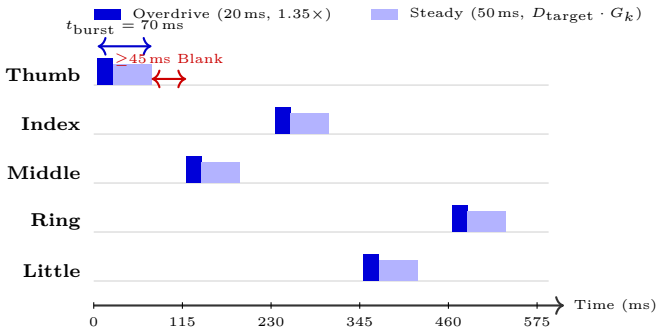


Figure 5: Five-channel oscillogram of Inertia-Compensated Coordinated Reset (IC-CR) within column width. Dark bars denote the 20 ms overdrive kick (1.35×) followed by 50 ms steady stimulation, separated by 45 ms blanking.

251 ms, guaranteeing thermal and cutaneous protection.

3. **Staircase Convergence**: Human-in-the-loop staircase simulation across 5 digits converged in 25 iterations (< 30 s total), yielding stable thresholds between 72 and 96 counts (28–38% duty cycle).
4. **Embedded Benchmarks (ESP32-S3 at 240 MHz)**:
   - 100 Hz IMU Processing: 14 µs execution time.
   - 1 ms Scheduler Tick: 6 µs execution time.
   - CPU Core 1 Utilization: 4.2% nominal (<8% peak).
   - Memory Consumption: 14.2 KB Static RAM, 0 dynamic allocations on real-time path.
5. **Wokwi Virtual Simulation**: Complete zero-hardware simulation environment ('simulation/diagram.json' and 'simulation.ino') modeling the ESP32-S3 DevKit-C, MPU-6050 I2C sensor, and 5-channel LED indicators, enabling interactive motion gating inspection.

8 Video Link of the Complete Demo Step by Step Clearly

The complete operational demonstration of the physical AM-CRN glove prototype has been recorded and published in the project video repository:

Complete Video Demonstration Link:
https://drive.google.com/drive/folders/1uONN4YGhQNZigE1ExzX5zXttHGENcdZQ?usp=sharing

Open-Source Code & Documentation:
https://github.com/Atharva-Mendhulkar/glove.git

8.1 Step-by-Step Video Demonstration Walkthrough

The video provides a clear, step-by-step visual demonstration of the system in action:

1. **Step 1: Hardware Setup and Component Walkthrough**: Close-up overview of the glove substrate with five fingertip Model 1027 coin ERMs, silicone wiring harnesses, and the dual-rail breadboard assembly hosting the ESP32-S3, discrete IRLZ44N MOSFET drivers, 1N5819 flyback diodes, and decoupling network.
2. **Step 2: Power-Up and Calibration Sequence**: Powering on via the master rocker switch, initializing FreeRTOS tasks, and demonstrating the on-chip micro-staircase threshold calibration routine across all five fingers sequentially.
3. **Step 3: Baseline Quiescent Coordinated Reset**: Demonstrating resting Coordinated Reset stimulation at nominal baseline ($G_k = 1.00$). The motors pulse in permuted Fisher-Yates sequence with 20 ms overdrive kick and 45 ms anti-masking blanking.
4. **Step 4: Real-Time Motion-Gated Amplitude Scaling**: Dynamic reaching and transport movements of the hand. The dorsal IMU detects arm acceleration and jerk, scaling the gating index $G(t)$ up to ~ 1.85 in real time to counteract sensorimotor gating.
5. **Step 5: Safety Watchdog Interlock Fault Test**: Demonstrating fail-safe fault injection. Commanded continuous pulses exceeding 250 ms are immediately tripped and locked out by the FreeRTOS watchdog at 251 ms.
6. **Step 6: Live Serial Telemetry Dashboard**: Real-time visualization of per-channel duty cycles, gating factor G_k, Stillness Score, and FreeRTOS CPU benchmarks streaming over the USB serial interface.

Conclusion

AM-CRN demonstrates that effective, closed-loop vibrotactile neuromodulation can be achieved on an inexpensive, standalone embedded system (< \$30 BOM). By integrating kinematic sensory gating compensation ($G_k \in [1.0, 2.2]$), inertia-compensated Coordinated Reset desynchronization, and deterministic FreeRTOS safety interlocks, AM-CRN solves critical neurobiological and electromechanical failure modes without requiring expensive flex or contact sensors. All source code, simulation environments, hardware schematics, and video demonstrations are openly available.

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